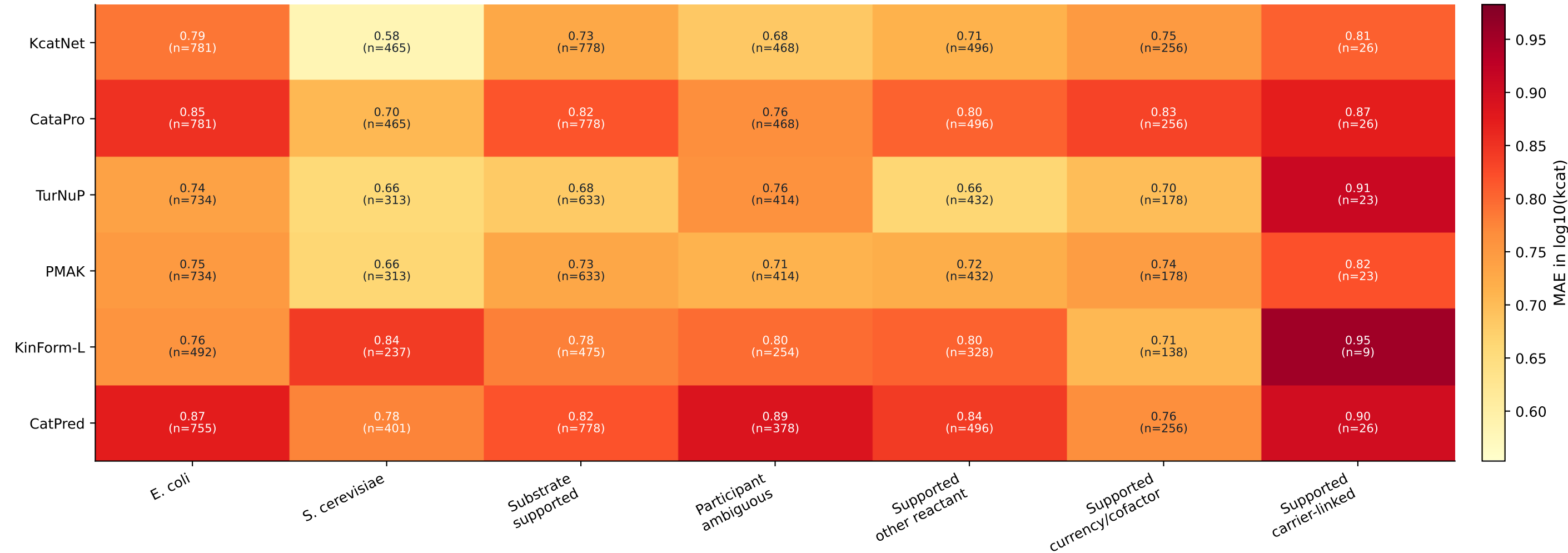
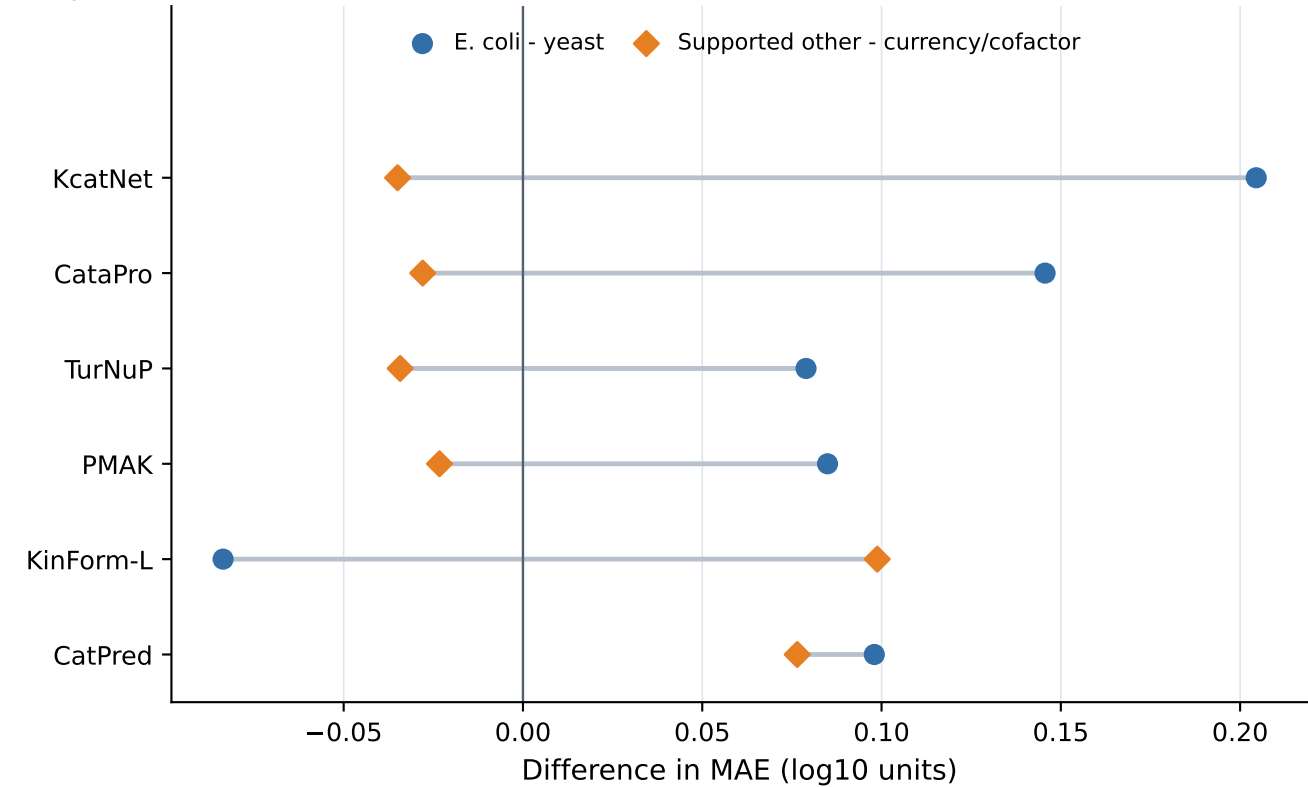


a Mean absolute error by species, substrate evidence and registry role



Role columns use only substrate-supported records and the joint name/identifier/standardized-structure registry; carrier-linked metabolites are separate.

b Subgroup MAE contrasts



c Large-error frequency by species

